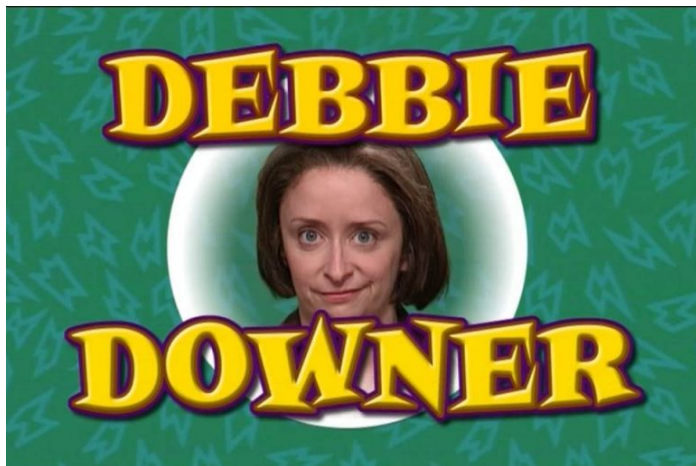


Regression assumptions and data cleaning

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Data cleaning

Regression assumptions

Influential observations

Multicollinearity

“90% of statistics is data cleaning.”

— Name missing



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- For categorical data with missing variables, can simply recode missing values as a special “unknown” category
- No easy solutions!

We'll use the college data set to predict a school's graduation rate from various factors. Take a look at the data; what potential issues do you see? We'll be using the following variables:

- Graduation rate
- Acceptance rate
- SAT score variables
- In-state tuition
- Out-of-state tuition

Many colleges have no SAT scores reported, so let's ignore those colleges (to enable a fair comparison) and also remove colleges with an obviously incorrect graduation rate of $> 100\%$:

```
my.sample <- colleges %>%  
  filter(!is.na(Average.combined.SAT) &  
         Graduation.rate <= 100)
```

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```

model <- lm(Graduation.rate ~ Average.combined.SAT + In.state.tuition,
            data=my.sample)
summary(model)

Call:
lm(formula = Graduation.rate ~ Average.combined.SAT + In.state.tuition,
    data = my.sample)

Residuals:
    Min       1Q   Median       3Q      Max
-45.526  -9.182   0.051   8.704  43.661

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    -8.3246456   4.3708279  -1.905   0.0572 .
Average.combined.SAT  0.0611221  0.0048878  12.505  <2e-16 ***
In.state.tuition    0.0012486  0.0001111  11.237  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.75 on 709 degrees of freedom
(19 observations deleted due to missingness)
Multiple R-squared:  0.4469, Adjusted R-squared:  0.4453
F-statistic: 286.4 on 2 and 709 DF,  p-value: < 2.2e-16

```



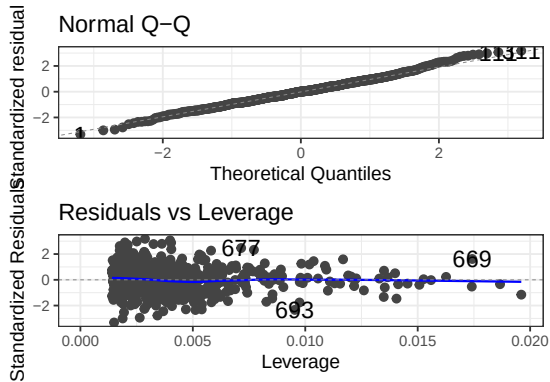
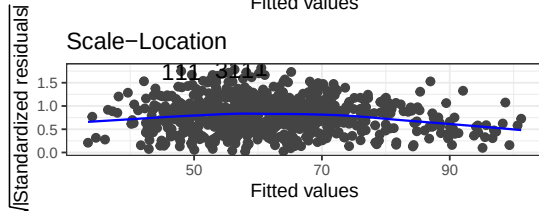
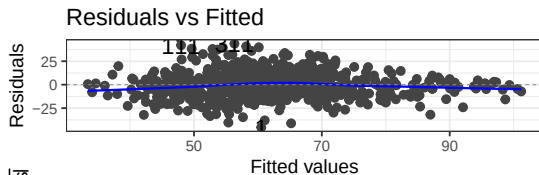
We need four things to be true for statistical inference (i.e., hypothesis tests, p -values, confidence intervals) to work for multiple regression:



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1. The errors are independent.
2. Y is a linear function of the X 's (except for the errors).
3. The errors are normally distributed.
4. The variance of Y is the same for any value of X ("homoscedasticity").

```
library(ggfortify)
autoplot(model) + theme_bw(base_size=9)
```





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- Independence is usually most problematic with time series data (i.e., $X = \text{time}$)
- Since each college is completely separate, there is no reason to think the errors are not independent
- However, we could have a violation of independence if e.g. all of the colleges in Texas (say) all implemented the same policies to try to improve graduation rates

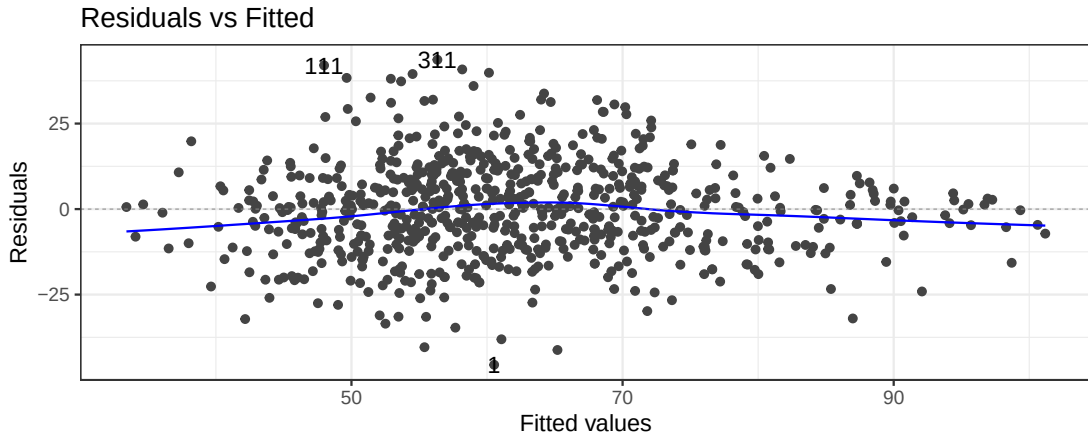


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Assumption 2: Linearity



Look at the residual plot—there should be **no trend** (the blue line should be roughly horizontal):



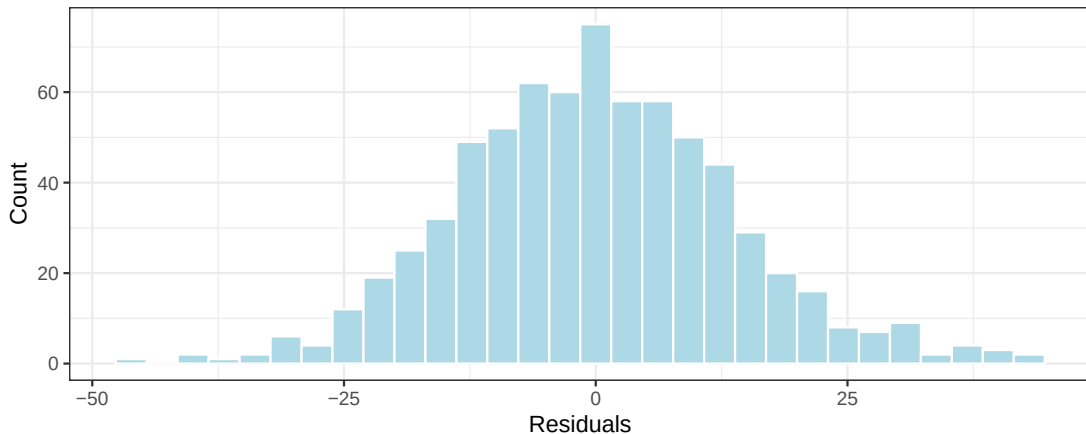


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Assumption 3: Normality of residuals



We need the histograms to be Normally distributed:

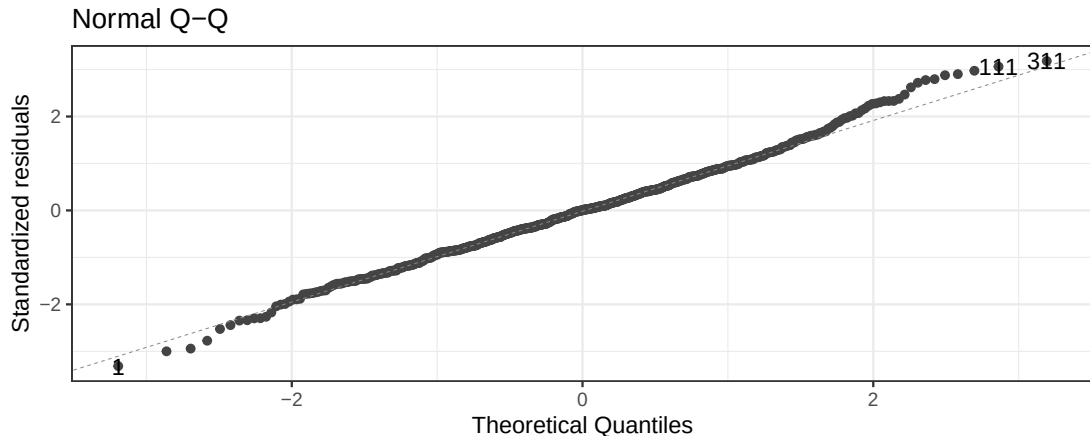


But it's hard to tell from a histogram!

Assumption 3: Normality of residuals



To be more careful we can use a Q-Q plot (a straight line indicates normality):



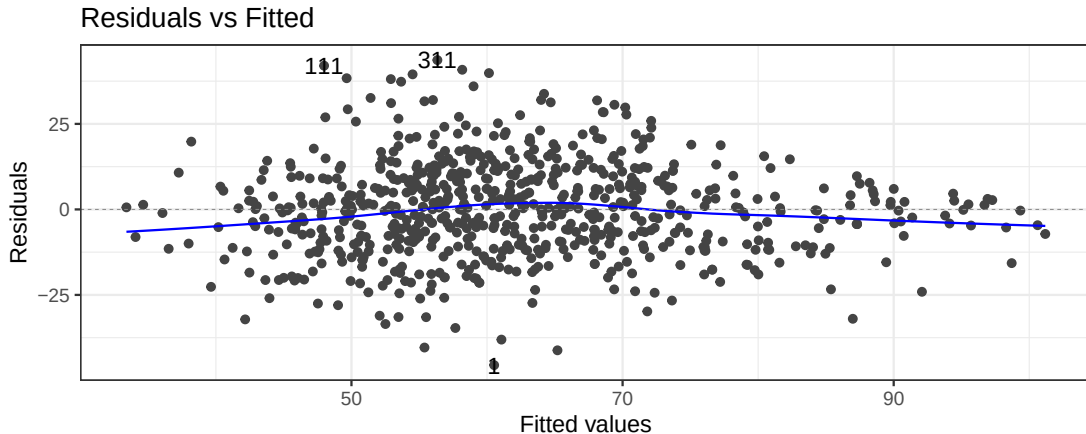


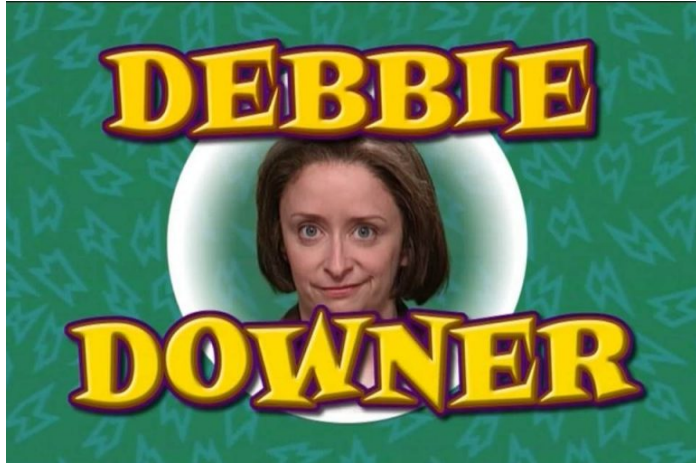
1. The errors are independent. ✓
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Assumption 4: Homoscedasticity



Look at the residual plot—we want a roughly constant vertical “thickness”:







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2. Y is a linear function of the X 's (except for the errors). ✓
3. The errors are normally distributed. ✓
4. The variance of Y is the same for any value of X ("homoscedasticity"). ✗



- If any assumption is not satisfied, we should not trust the p -values or confidence intervals that come out of the model



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- However, as long as linearity is satisfied, we can still use the model for making predictions (we just can't put reliable CIs on those predictions)
- Is there anything else we can do? (Yes—stay tuned for next week!)

Data cleaning

Regression assumptions

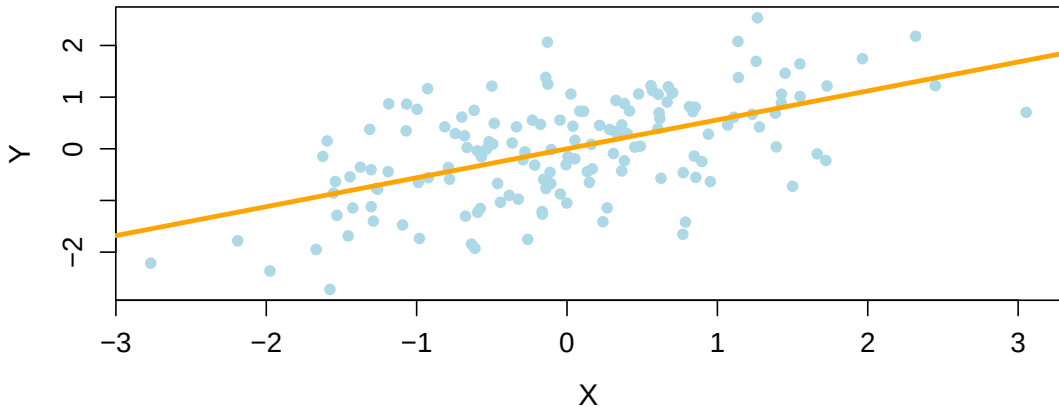
Influential observations

Multicollinearity

What a single case can do

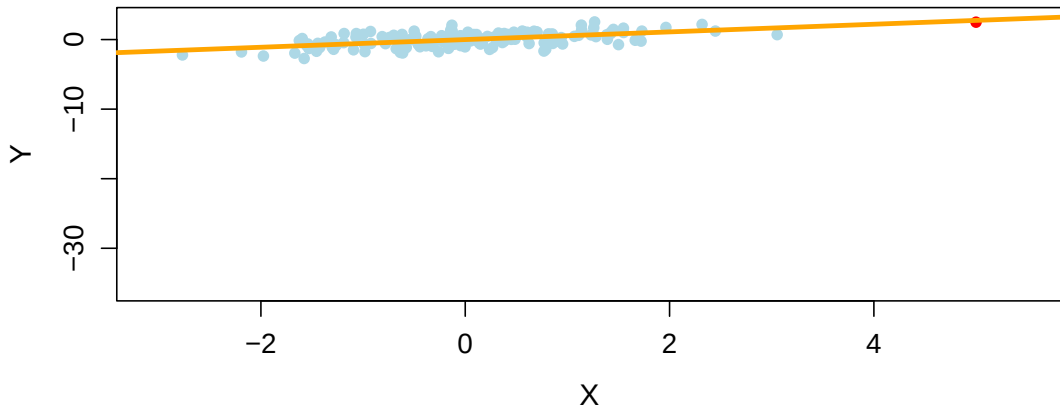


Let's take some hypothetical sample data:

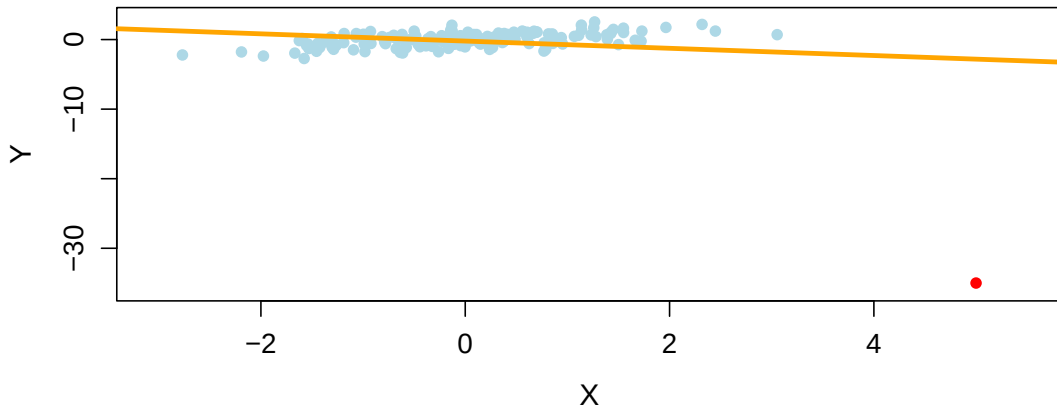


Now add a single outlier at $X = 5$ and watch the line.

What a single case can do



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Blackmail:

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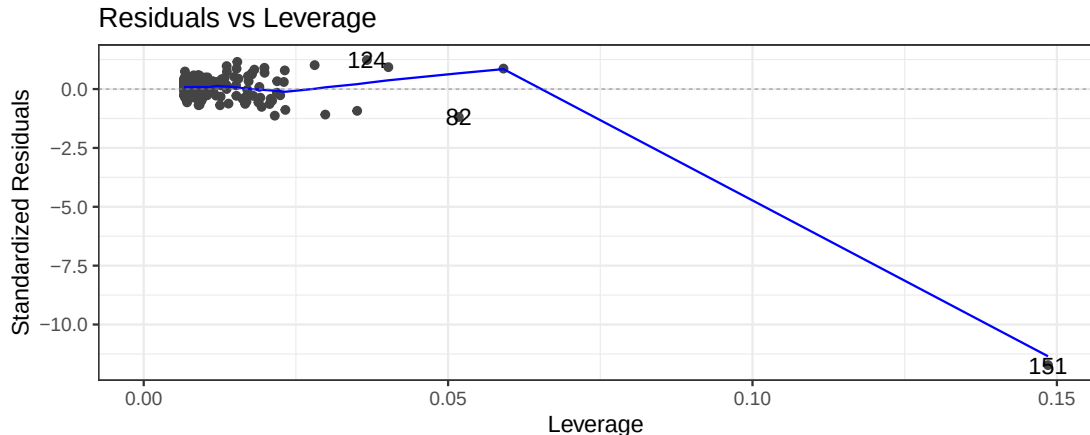
Regression:

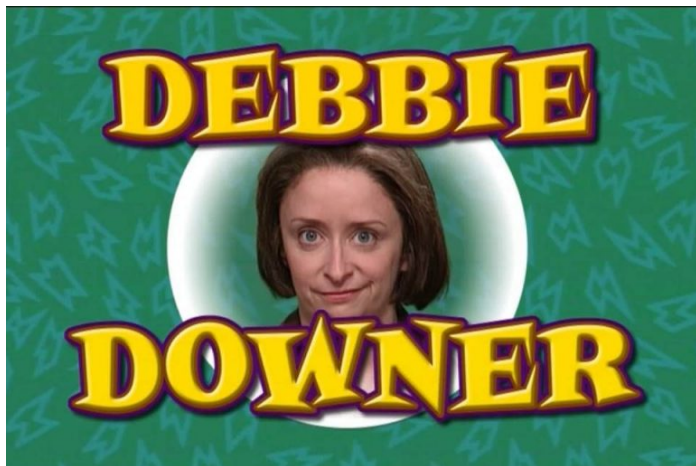
- When a case has a very unusual X value (i.e., far from $\overline{X}$), it has **leverage**—the *potential* to have a big impact on the regression line
- When that case *also* has a Y value that is out of line with the general trend, it will pull the regression line towards it—giving it **influence**

How do I know what cases are influential?



Look for cases with high leverage and a large (positive or negative) residual:







- If removing the influential observation doesn't make a substantial difference in your analysis, don't worry about it
- If it does:
 - Consider whether it could be a mistake (happens more than you might think!); if it is, correct the error or drop the case
 - If not, hold out the influential observation(s) and report on them separately
 - Do not just throw out and ignore influential observations!

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- Hard to estimate the effect of changing X_1 while holding X_2 fixed, so standard errors and p-values blow up – and the coefficients are **unstable**.
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Correlation between the response and the predictors is good. Correlation *among* the predictors is not!

How can we detect multicollinearity?



For two predictors, check their correlation:

```
cor(my.sample$Average.math.SAT,  
     my.sample$Average.verbal.SAT,  
     use="complete.obs")
```

```
[1] 0.9194207
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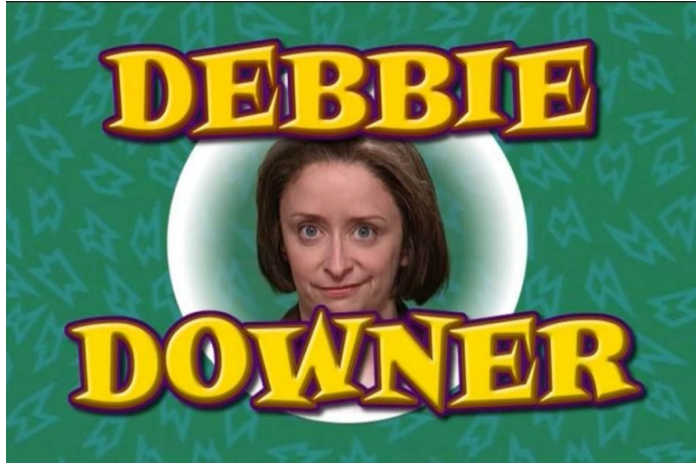
```
[1] 0.9194207
```

For 3+ predictors, use the **variance inflation factor**, $VIF(\beta_j) = 1/(1 - R_j^2)$, where R_j^2 comes from regressing X_j on the other predictors. It is 1 when X_j is unrelated to the others and ∞ under perfect multicollinearity.

```
model <- lm(Graduation.rate ~ Average.math.SAT +  
            Average.verbal.SAT + Acceptance.rate,  
            data=my.sample)  
library(car)  
vif(model)
```

Average.math.SAT	Average.verbal.SAT	Acceptance.rate
6.647073	6.454614	1.224408

Predictors with $VIF > 5$ are a cause for concern.





- **Drop** one of the high-VIF variables and refit. (Just like dropping one dummy when putting a categorical variable in the model!)
- **Combine** the correlated variables into a composite. (Combined SAT = Math + Verbal)
- **Accept it** – collinearity is not always a problem: when the collinear X 's are only there for adjustment and the coefficient you care about has low VIF; when it comes from how we build X (polynomial terms – next week! – dummies, interactions); or when you only want to predict, at typical X values.